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The First Fifteen Years of Computing in Aotearoa New Zealand

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Abstract—This article outlines the first fifteen years of modern computing in New Zealand. We discuss the growth in computers and staff, types of usage including service bureaus, the professionalization of the industry, and the corresponding educational and academic developments. We describe local innovations, and then the economic and social impact of computing, during a period of major economic and social change for the country as it decoupled from its historical dependency on the United Kingdom.

Modern computers arrived in Aotearoa New Zealand in 1960, less than a century after the first discernible information technology [25]. New Zealand was then a relatively isolated and small economy (with a growing population of about 2.5 million), so it is feasible to track the dissemination and socio-economic influence of computing for the following period; this study runs from 1960 to about 1975. Two events in the mid-1970s signaled the end of the economic and social era that began after World War II, and act as the principal end points of our study: Britain joined the European Common Market in 1973, with major economic consequences, and New Zealand switched on a national Police Computer in 1976, with significant social impact.

In 1960, New Zealand had a rather centrally managed economy. Indeed, its currency did not float until as late as 1985. The details are complex [44], but the result was that during the period of our study the Treasury was always concerned about the foreign exchange impact of computer imports, and this necessitated a system of import licensing. This was a constant background for trends in computing, especially since the Treasury initially favored computing service bureaus to ensure maximum usage of what they considered a scarce resource. Of course, many companies much preferred to have their own systems.

Several themes are used to organize this article:

- growth in numbers of computers, vendors and employment
- key areas and types of usage, including service bureaus
- professionalization
- training, teaching and academia
- local contributions to the technology
- commercial and economic impact
- social impact

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GROWTH IN NUMBERS AND EMPLOYMENT

Table 1 shows the estimated number of electronic digital computers installed in New Zealand over the years in question.

Year	Computers	Source	Notes
1960	2	[24]	
1965	70	[21]	[32] gives 45
1966	81	[21]	
1967	100	[3]	
1968	120	[21]	[12] gives 87
1969	140	[21]	
1971	180	[21]	
1972	200	[12]	
1974	280	[21]	
1976	400	[21]	

TABLE 1. Estimated number of electronic digital computers installed in New Zealand.

A snapshot of installations in Auckland alone lists 37 sites at the end of 1969, covering businesses of many sorts, local government, a university, and even the Naval Research Laboratory [20]. At least seven of

FIGURE 1. Fletcher Computing Bureau staff, 1966. Courtesy Fletcher Archives, NZ.

the sites appear to be service bureaus. The author noted in passing that there was also a large Government investment in computers in Wellington.

Although there are some discrepancies in the available data, the trend is one of rapid growth, from one computer for every 1.2 million people to one for every 8000. Similarly, the number of programmers in public services rose from a handful (maybe 4) in 1960 to 115 (84 men and 31 women) by 1974 [12]. Overall, some 4000 data processing staff were then employed at 220 sites – 39% in Wellington, 31% in Auckland, 9% elsewhere in the North Island, and 21% in the South Island [21]. Pay was attractive: in 1967, an EDP manager's average annual salary was NZ$4580 [3], when the national average for company directors and managers was NZ$4220 [5]. Programmers and systems analysts earned anywhere between NZ$2000 and NZ$4000.

Even if the public services had 27% of female staff, it is not clear that the same applied elsewhere. For example, a 1966 staff photograph of Fletcher Building's internal computer bureau shows 14 men and only three women (Figure 1). Whenever staff appear in computer room photographs, the women are almost invariably seated at a keyboard and the men are standing in discussion. A 1967 survey [3] shows systems design and programming as 90% male occupations, operations as 50% male, and data preparation as 97% female.

There is little to indicate significant Māori involvement in computing during our period of study. Robyn Kāmira [34] shows that such involvement would only start in the 1980s, when a first handful of Māori information technology specialists graduated.

New Zealand had a long history of preferring to import British products of all kinds, but this did not long survive in the computer market. IBM was of course a major player; not only did they snag the Treasury as their first customer for an IBM 650 in 1960 [24], but they also delivered a heavily discounted IBM 1620 to the University of Canterbury in 1962, the first machine in academia [16]. This was in line with IBM's world-wide practice of educational discounts, a powerful marketing tool that exposed many students to the IBM brand. As time went on, IBM sold numerous machines - more IBM 1620s, at least one IBM 1401, several IBM 1130s, until the game-changing IBM 360 series came along in 1964.

The main British competitor in 1960 was ICT, but a potential customer would certainly have been confused by the fragmented state of the British computer industry at that time [31]. Nevertheless, ICT sold one ICT 1201, at least six ICT 1301s, and numerous machines from the ICT 1900 series. It was not until the various British companies were unified as ICL in 1968 that things settled down, but by then IBM had the lion's share of the New Zealand market and ICL was falling behind. In 1974, the new ICL 2900 series had to compete with the well established and upwards compatible IBM 360 series.

Other companies that were active in the early days were Burroughs (with its E2000, not a general-purpose computer, but specialized for accounting), and NCR (Series 500, also specialized for ledgers).

We have collected available information about individual computers present in New Zealand during the period under study, but at best it is a small sample of the market. At the time of writing, we have identified more than 80 machines (including programmable calculators) with arrival dates up to 1974, but the estimated total for that year was 280. Many of the earlier machines had of course been scrapped by then, so our sample certainly covers less than 25% of the market. The distribution of this sample across manufacturers is shown in Table 2. The total for the British companies that ended up as part of ICL is 29, similar to the total for IBM.

Manufacturer	Count
Burroughs	12
DEC	9
HP	1
IBM	28
NCR	2
EELM	3
Elliot	1
ICL	8
ICT	17

TABLE 2. Sample of market distribution (US and UK vendors).

FIGURE 2. ICL 1902A delivery to Motor Specialties in 1969. Courtesy Fletcher Archives, NZ.

FIGURE 3. The TAB Computer Centre, Christchurch, 1970. Courtesy Fletcher Archives, NZ.

FIGURE 4. Ceramic House, 1969. Courtesy Auckland Libraries Heritage Collections.

Throughout the period under study, most computers remained large and comparatively heavy, even as they moved from “first generation” (vacuum tubes) through “second generation” (discrete transistors) and “third generation” (integrated circuits) to the “fourth generation” (very large scale integration). As the technology got smaller, mainframe computers became more powerful rather than more compact; space and air conditioning requirements did not substantially change. For example, the third computer installed in 1969 by Motor Specialties in Auckland, an ICL 1902A, needed to be lifted to the top floor by crane, like its predecessors, an ICT 1201 and an ICT 1301 (Figure 2). Only after 1970 when minicomputers such as the DEC PDP-11 entered the market did this change, so that some computers could be installed almost as easily as domestic appliances.

Originally, much equipment was rented; in 1967, 70% of “owners” actually rented their equipment, and a further 15% rented part of it [3]. By the late 1960s, the growth in installations led to a minor building boom of dedicated computer centers, which tended to be stark concrete buildings with few windows (Figure 3). A rare exception was Ceramic House in Auckland (Figure 4), specifically designed to accomodate an ICT 1901 delivered in 1969. This building was designated as a heritage site in 2024.

This section has described how computers were imported, housed, installed and staffed during the period of study. We will now consider what they were used for.

HOW COMPUTERS WERE USED

In 1960, the main expected workload was the payrolls for teachers and public servants. The expansion in fields of use came very quickly. By the end of 1966 [10] [**TBD**: Janet check the Archives citation], Government usage included payroll, statistics, science, NZ Railways, Inland Revenue, Post Office, Social Security, Reserve Bank, Bank of New Zealand, New Zealand Broadcasting Corporation, Apple & Pear Board, and the Dairy Board. In a surviving list of import license applications between August 1965 and May 1966 [10], there are three applications from local government, three from universities, but 30 from business and primary and secondary industries. The New Zealand economy took to computers rather quickly, held back mainly by import license restrictions.

Applications of importance for business users were listed in 1967 as invoicing, payroll, general ledger, stock control, expense analysis and sales statistics [3]. All the same, mathematical, scientific and project planning applications such as PERT (Program Evaluation and Review Technique) were increasingly used. Users saw the advantages of computerization as speed, accuracy, improved reporting, improved organizational control and staff savings. Quite surprisingly, in 1967 most computers outside of academia were underutilized, with 250 operating hours per month on average, and single-shift operation still being typical. Only 68% of installations charged internal users for usage: "one of the chief reasons why installations can be uneconomic".

In 1973, the Department of Statistics provided the following summary [8]:

> The demand for computers has come from Government departments, local authorities, universities, primary producer boards, private firms in industries such as printing, forestry, insurance, oil, food processing, electrical equipment manufacturing, building and construction, clothing, engineering, airways, banking, retailing, motor assembly, paint manufacturing, and stock and station agents.

Because of government's reluctance to issue import licenses, numerous service bureaus arose to allow resource-sharing. By 1967, when the total number of computers was about 100, at least 166 companies were using bureau services operated by computer vendors, and another 78 were using independent bureaus [3]. Table 3 lists some example bureaus and their approximates dates of foundation, but the list is far from complete.

Year	Name	Location	Notes
1961	Electronic Data Processing Ltd	Auckland	Closed 1963
1963	Electronic Data Systems Ltd	Wellington, later Auckland & Christchurch	50% EELM, 50% NZ Truth (until 1968)
1965	Computer Bureau Ltd	Christchurch, later nationwide	Renamed Datacom
1966	Allied Computer Processors	Dunedin	
1967	Databank Systems Ltd	multiple	Banking consortium
1968	Electronic Data Processing NZ Ltd	Wellington	
1968	NCR Bureau	Wellington	
1968	Burroughs Bureau	Wellington	
1968	ICL Centre	Auckland	
1969	Computer Activities	Auckland	
1969	Computer Systems	Auckland	
1969	Data Enterprises Ltd	Tauranga	Paul and David Walker

TABLE 3. Example computer bureaus in New Zealand.

The first bureau listed above remains something of an enigma. According to ICT's records in the U.K., it bought an ICT 1301 for "service work"; but we have found no local records to substantiate this, beyond the formal records of the formation and dissolution of the company. Another phantom bureau hosted the government Education Department's second computer, also an ICT 1301, leased from ICT at the end of 1963 and installed in ICT premises in Wellington. When the Minister of Education pressed a button to inaugurate it, it was said to be "New Zealand's first computer bureau" [2], although all indications are that it was dedicated to the Education Department. It seems that obtaining an import license for a shared bureau computer was significantly easier than for a single-user machine. The other bureaus were real enough, and we now present a few brief case studies.

Electronic Data Systems Ltd (EDS), founded in 1963, was jointly owned by the British company English Electric Leo Marconi (EELM) and the NZ Truth newspaper. It had no connection with the later American company of the same name. EELM was one of the numerous firms that ended up as part of ICL, and presumably wanted to enter the New Zealand market. Apart from its earlier DEUCE and LEO computers, it made a family of KDF machines, including the KDF6, of which at least 12 were sold [31]. The KDF6 sales records are scanty [17], but EDS had such machines in Wellington and Auckland [30]. Indeed we have photographic evidence of the machine in Auckland (Figure 5).

NZ Truth, described by the National Library web site as "one of the country's most colourful, controversial and popular newspapers" ran from 1905 until 2007. It is unclear why it decided to enter the computer bureau business in 1963. In any case, EDS opened a third branch in Christchurch in 1970, and in due course switched to Burroughs and ICL equipment [7]. By 1972, the Christchurch and Wellington offices were linked by a remote job entry system. EDS remained in business in one form or another until 1994, decades after EELM no longer existed and NZ Truth had sold its stake.

Another success story started in 1965. G. Bernard Battersby (1925-1997) was the Dean of the Faculty of Commerce at the University of Canterbury, and a government advisor. Yet he joined with Paul M.

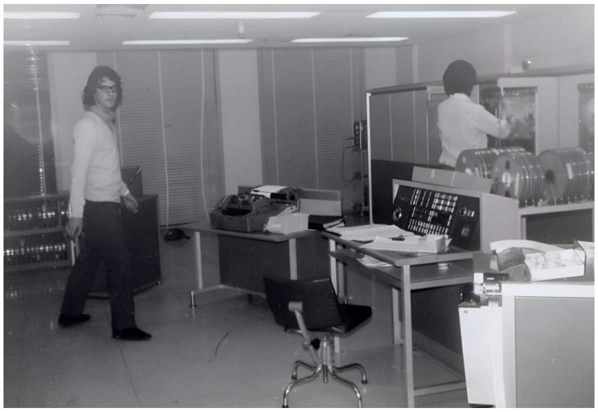

FIGURE 5. EDS KDF6, Princes St, Auckland in about 1970. Courtesy Neil Harsant.

Hargreaves (1938-2014) to found Computer Bureau Ltd (CBL) in Christchurch [14]. Its first computer was an ICT 1902 delivered in 1966; with a staff of seven, CBL was unprofitable until 1969, when a branch was opened in Wellington. In early 1970, CBL had some 50 customers, and the Christchurch computer was working 140 hours per week. Thus an ICL 1902A was delivered to Christchurch, the orginal 1902 was moved to Wellington, and an additional ICL 1902 was ordered for another new branch in Hamilton (Figure 6). In Auckland, CBL had taken over Fletcher Building's in-house computer bureau, also ICL-based. All training of systems analysts and programmers took place in Christchurch, with total staff expected to reach 70 during 1970 [4]. By 1974, CBL needed to order three ICL 1902T models for Auckland, Wellington and Christchurch, with a nationwide total of six computer centers and 125 staff. Furthermore, they told the newspapers [9]:

> [CBL] is unique in that many of its major users are also shareholders, a number of whom previously operated their own inhouse computers. These include Fletcher Holdings, Ltd, Wilson and Horton, Ltd, the New Zealand Co-op Dairy Company, Bing Harris Sargood, Ltd, United Dominion Corporation, Ltd, and in Christchurch, the Canterbury Building Society, the S.I.M.U. Mutual Insurance Association, and the City Council.

Clearly, Computer Bureau Ltd had found a winning formula. In 2025, they still prosper under the name of Datacom, claimed to be the largest technology company in New Zealand, employing at least 6000 people at offices across the Asia-Pacific region (Figure 7).

In the early 1960s the activities of New Zealand's

FIGURE 6. Computer Bureau delivery in Hamilton, 1971. Courtesy Fletcher Archives, NZ.

FIGURE 7. Datacom, Auckland, 2025. Courtesy Brian Carpenter.

trading banks were heavily restricted by government regulations. Trading banks were not allowed to compete directly with the Post Office Savings Bank or Building Societies and as a result their importance in the financial system decreased, falling from 63% of total deposits in 1934 to 22% in 1962. In response to this challenging situation, the five trading banks adopted a collaborative approach: pooling their resources to buy computers together and to manage their computing services. Databank Systems Limited was set up in 1967 as a private company with a share capital of NZ$200,000. Initially Databank's main objectives were:

1) To automate bank services;
2) To provide trading banks with a management information system;
3) To provide bureau services for parties other than banks.

By August 1968, Databank was operating from four centers in Auckland, Wellington, Hamilton and Christchurch. By 1969, two further centers were opened at Palmerston North and Dunedin. By November 1969, every branch of the five trading banks in the country had been converted to computer processing. Any concerns that competition would be compromised was quickly dispelled by the expansion of banking services to all participating banks. By 1972, Databank was reporting a healthy return of NZ$49,974 to shareholders, and a transaction revenue of NZ$103.7 million.

When Databank was founded it had a staff of 84, but by 1977, it had over 1000 staff making it one of the largest employers in the country. By that time it was processing an average of one and a half million bank transactions per day and servicing over a thousand computer programs. The number of bank master file records was over ten million. Databank became the largest data processing company in the Southern Hemisphere and played a significant role in New Zealand's economic development.

However, problems started as early as 1970 when Databank's charismatic General Manager, Gordon Hogg attempted to open up Databank by providing bureau services to outside customers. This met with resistance from the banks, who preferred Databank to stay low profile. They were concerned that customers would feel their personal information had been compromised if they became aware it was being processed by an organization separate from their own bank. The banks had taken a bold and unusual step by cooperating but were not prepared to extend this cooperative approach to organizations beyond their own sector. This resulted in ongoing disagreements between the entrepreneurial General Manager, who saw the opportunities bureau computing could offer, and the cautious and risk-averse banking sector.

Databank was successful during its early years, as the banks were in a period of weak competition and strong co-operation. At the time it was established there was limited competition as banking was highly regulated. However by the time the Company was sold in 1994, this situation had reversed. In 1984, New Zealand adopted a neoliberal economic approach and banks were now competing strongly for customers, cooperation was weak and competition was high. In 1994, Databank was sold to the Texas-based company EDS for an estimated NZ$100 million.

The highly regulated banking sector in New Zealand left the trading banks with few options to increase their shrinking share of the financial market. The new technology of computing offered an opportunity and due to the Government's policy of insulationism and restrictions on import licenses of overseas technology the only practical way they banks could make use of this was through collaboration [45], [19].

Thus, as computer usage spread throughout business and government, import license pressure encouraged resource sharing and successful bureau operations. This pressure was not to be relieved until major economic reforms in the 1980s, outside our study period.

PROFESSIONALIZATION AND ACADEMIA

The New Zealand Computer Society

Professionalization started early. The New Zealand Computer Society (NZCS) was founded in the same year that the first modern computers arrived in the country. Originally known as The Data Processing and Computer Society, NZCS was registered on 6 October 1960. Initial members were enthusiasts from the academic and business communities, who were aware of the impact computing was having overseas. In the early days academics had the stronger voice and there were restrictions on the numbers of members from the "machine houses" due to an unfounded fear that they would use the Society for promotion rather than learning. The Computer Society had two complementary roles. The first was to look after the interests of its members; the second involved meeting a broader obligation to society in general. The Society wanted to be "the independent and authoritative source of information on matters to do with computing and its social impact" . This was attractive since self-regulation could be a way of forestalling legislative constraints.

Wellington (the capital) was the first branch, followed by Canterbury, Otago and Auckland. Initial activity was very much around the branch level. By 1965, there were over 200 members and the possibilty of holding national conferences and publishing a journal was first raised. In 1968, the Society organised its first national conference, which was so well attended that it became a regular occurrence. As well as the presentation of papers, this conference was an important opportunity for computer vendors to present their wares. Around 500 people, on average, attended each of the conferences. An overview of the first five conferences is provided in Table 4. As well as local keynote speakers, well known figures in the computing world from the USA, the UK and Australia also spoke at NZCS conferences. For example, at the 1970 conference, Richard Hamming from Bell Telephones, USA spoke on "Computers and Computing in the 1970s", in 1972 Paul Armer from Stanford University covered "Computers in Society" and in 1976, ACM President Jean Sammet spoke on "Responsibilities in Computing Professionals". These keynotes were a major draw at conferences and widely reported on in the local media.

Year	Location	Theme	Papers
1968	Auckland	No theme	24
1970	Dunedin	No theme	35
1972	Palmerston North	Computers in the Community	32
1974	Christchurch	Practical Computing	20
1976	Hamilton	Responsibility	19

TABLE 4. Early NZCS conferences.

By 1970, the NZCS was accepted as the voice of computer practitioners in the country by both government and the public. That recognition came with responsibilities and the Society started to examine social issues around the use of computing, as well as the quality of computing education on offer. A Certificate in Data Processing was set up, and the Society also developed a Code of Ethics. The foundation President of NZCS (1960-1967), Professor Gordon Oed, is an excellent example of the links between early computing, academia and the business world. He was a senior lecturer in accounting at Victoria University of Wellington, and formerly a chair of the Wellington branch of accountants. Dr Bernard Battersby who has been previously mentioned as the co-founder of Datacom (formerly Computer Bureau Ltd.), was also President from 1967-1969. His profile notes that Datacom had to wait until his term as President was over before making its first profit.

Later, NZCS rebranded itself as Information Technology Professionals New Zealand (ITPNZ) [35], and eventually underwent a severe financial crisis during the drafting of this article.

Training and Academia

Naturally, the need for training was an issue from a very early stage. By 1967, it was a serious concern: "It is not sufficient for the Universities to provide computer facilities and training only at the scientific level. The Technical and Management Institutes cannot be left to their own devices... the secondary schools (and before too long the primary schools) must introduce computer knowledge" [3]. In practice, most training originally came from computer vendors, or people learned on the job. The first NZCS national conference in 1968 led to a committee on training requirements, which recommended computer science undergraduate courses at every university, as well as some postgrad and extramural courses. The committee "saw it as the primary role of Computer Science departments to educate system programmers, as well as future researchers" and recommended that "Commerce Departments should deal with teaching system analysts and managers" [15].

According to Tim Bell [22], the first complete tertiary course was offered by the Auckland Technical Institute (ATI, now Auckland University of Technology) in 1967, teaching Fortran, RPG, and COBOL, along with mathematics, statistics, and accounting. Although ATI had acquired a third-hand ICT 1201 in 1966 [24], student programs were processed overnight by a local company. By 1970, various technical colleges were offering courses in computing and programming, but with only 46 students enrolled nationally [8].

As universities acquired their own computers from 1963 onwards, they naturally started offering courses on the use of computers for mathematics, statistics, science in general, and business. Often, such courses were taught by computer center staff rather than by academic faculty. Some teaching of computing in secondary schools started from about 1974, but Bell [22] wrote that "finding computers that students could program was a challenge, and there was no national initiative around computers in schools until the end of the 20th century." Indeed, Massey University was offering extramural computer science courses by the mid-1970s, and the most eager participants were secondary school teachers.

Among degree-granting institutions, the University of Canterbury was the first to acquire a computer, in 1962. The following table lists the first computers at each university.

Year	University	Model	Notes
1962	Canterbury, Christchurch	IBM 1620	
1963	Auckland	IBM 1620†	
1965	Victoria, Wellington	Elliot 503	Shared with DSIR
1965	Massey, Palmerston North	IBM 1620	
1966	Otago, Dunedin	IBM 360/30	
1969	Waikato, Hamilton	IBM 1130†	

Various other machines were purchased, culminating in a joint purchase of five Burroughs B6700 mainframes for Auckland, Massey, Wellington, Canterbury and Otago in 1971 [27]. (Waikato and Canterbury's Lincoln College campus were fobbed off with underperforming remote access to Auckland and Canterbury respectively.) The original machines were primarily used in support of research in various disciplines; administrative use came years later. For example, in Auckland the IBM 1620, and an IBM 1130 that joined it in 1967, were much used by economists, who were "often the heaviest user," running simulations of econometric models in 1964-70. Applications, written in Fortran, included dynamic linear models of the New Zealand economy, dynamic simultaneous equation models, and systems of linear stochastic differential equations [40]. The researchers were students and colleagues of Professor Rex Bergstrom, who had cut his teeth on the Cambridge EDSAC in 1952-54. Of course, mathematicians, scientists, and engineers were also early users.

At Canterbury, the IBM 1620 was soon "heavily used by a wide range of university departments, frequently logging more than 500 hours a month... By July 1964, after 2 years of use, the University's computer had been used in more than 120 projects by 14 departments, and more than three-quarters of these projects could not have been tackled without a computer" [16].

As well as research, these early machines were also used for undergraduate programming courses. However, in the 1960s, computing was mainly taught as a useful skill, not as an academic subject in its own right. The discipline of Computer Science (CS) was late to arrive in New Zealand. For example, in the UK (the largest source of academic immigrants at that time), the Mathematical Laboratory in Cambridge had offered a post-graduate Diploma in Numerical Analysis and Automatic Computing (later renamed Diploma in Computer Science) since 1953, and the Department of Computer Science at the University of Manchester was founded in 1964, graduating its first students in 1968. CS departments also emerged in the USA during the 1960s. It was not until 1973 that three CS departments were founded in New Zealand, at the University of Canterbury (Professor John Penny), Massey University (Professor Graham Tate) and Waikato University (Professor Darryl Smith) [22], [18]. CS departments at Auckland, Wellington and Otago were not established until the 1980s, outside our study period. In Auckland at least, this reluctance was caused partly by considerations about duplication of effort, but also by assertions that Computer Science was not a "proper" science - one view was even that computing was merely a branch of physics [23]. At Otago, Brian Cox was appointed as "Lecturer in Charge of the Computer centre" in 1964, even though the first computer, an IBM 360/30, did not arrive until 1966. Cox had used the EDSAC 2 in Cambridge as a physics graduate student, trained by Maurice Wilkes and David Barron. Although computing courses were taught from 1966, and Computer Science papers from 1972, Cox had to wait until 1984 to become the founding Professor of Computer Science [15].

† This is preserved at the Bob Doran Museum of Computing.

LOCAL CONTRIBUTIONS

Unlike Australia, there appear to have been no early attempts to build a complete digital computer in New Zealand. However, during the 1960s and 1970s, computing in New Zealand's universities was characterized by adaptation and experimentation. Limited access to imported hardware, combined with a shortage of technical support and software, encouraged a culture of innovation. Systems were modified, extended, and repurposed to meet local research and teaching needs, reflecting both necessity and ingenuity. Most of these efforts are known only by anecdotal evidence.

The IBM 1620 and Early Student Experimentation

At the University of Auckland, the IBM 1620 served as the principal computing resource for the entire campus. Access was tightly rationed through a booking system, and graduate students frequently worked overnight to obtain machine time. The environment fostered both technical creativity and practical problem solving.

Recognizing the need to monitor unattended batch processing, Brian Hicks and Murray Johns of the Physics Department designed a simple alarm system that detected when the 1620 halted during execution. Photocells placed over the Read No Feed and Check Stop indicator lights triggered an audible alert if either light remained active, allowing operators to intervene promptly. This “computer watcher” represents one of the earliest examples of locally developed operational aids for computing in New Zealand.

The IBM 1130 and Hardware–Software Adaptation

The subsequent installation of an IBM 1130 marked a transition to more versatile, interactive computing. Although the system was modestly configured with 8 KB of storage, it became the focus of significant technical development. Brian Hicks added a 1440 chain printer through a locally engineered Storage Access Channel (SAC) interface, while Peter Fenwick constructed a floating-point arithmetic unit to accelerate numerical computation. The latter experiment was eventually abandoned due to the unreliability of the 8-input OR gates used in its logic design.

Software enhancements complemented these hardware efforts. Murray Johns modified IBM’s standard operating system to implement an integer-only execution mode, improving performance by an order of magnitude for users not requiring floating-point precision. Nevil Brownlee, who became Computer Manager in 1970, developed a real-time clock for the 1130, subsequently replicated at Massey University. At the University of Waikato, by contrast, the same model was operated in its standard configuration, without local modifications.

Databank and the Commercial Mainframe Environment

While universities pursued their own adaptations, the commercial sector was also expanding its computational capacity. In 1966, IBM installed two System/360 Model 30 systems for the Bank of New Zealand—one in Wellington and one in Auckland—to support Databank, as described above. Each system possessed 16 KB of ferrite-core memory and five magnetic tape drives.

The configuration evolved rapidly. Tape-based storage was replaced by 2311 and later 2314 disk drives; 1419 MICR readers processed up to 1 600 cheques per minute; and by 1969 Databank had expanded to six regional centers operating on System/360 Model 40 systems equipped with 128 KB of memory. This initiative represented one of the earliest examples of distributed commercial data processing in New Zealand.

In 1968, Progeni, widely regarded as New Zealand’s first software house, was founded by Perce Harpham [29]. Its establishment signalled a broader transition from predominantly academic and experimental computing to an emerging commercial software industry. Later, Progeni was to form a joint venture, Polycorp, in 1981 to develop and market the Poly 1—one of the most successful computers designed and manufactured in New Zealand [26].

The Burroughs B6700 and Standardization in Higher Education

As mentioned above, in the early 1970s, the University Grants Committee adopted a policy of standardizing university computing on the Burroughs B6700 system. Auckland, Massey, Victoria, Canterbury, and Otago each received a B6700 installation, while Waikato—then a smaller and newer institution—was provided with a Remote Job Entry (RJE) terminal to submit punched-card jobs to Auckland.

Nevil Brownlee edited the *Journal for Users of the B6700* (JUB), a communication medium for sharing technical notes and programming techniques among New Zealand users.

At Massey University, the need for interactive computing led to further innovation by 1975. Burroughs block-mode terminals were prohibitively expensive, so the Computer Unit acquired a DEC PDP-11/10 to function as a terminal concentrator for low-cost VT05 terminals. Brian Carpenter ported CERN’s PL-11 compiler to the B6700 environment, and Ted Drawneek developed the necessary system software. This hybrid solution extended interactive computing capabilities well beyond those supported by the Burroughs hardware alone.

Early Computing Research and Algorithmic Innovation

Local innovation also extended to scientific programming. In 1956, John Butcher—later Professor of Mathematics at Auckland, and a founding member and the first head of the Auckland CS department—was among the earliest New Zealanders to undertake computational research overseas, running simulations of cosmic-ray showers on the Australian-built SILLIAC (a version of the ILLIAC). In doing so, he devised an efficient recursive method for managing intermediate results, effectively anticipating the modern concept of a software stack. His approach reduced storage requirements to the logarithm of the problem size, exemplifying the algorithmic economy required when computing resources were severely constrained.

Programming Languages

In 1966, Bruce Moon, then at the University of Canterbury, published the first locally written programming textbook [36], marking a milestone in the development of computing education in New Zealand.

Within the University of Auckland's Computer Science Department, a locally designed programming language known as SMALL (Small Machine Algol-Like Language) was developed under Nevil Brownlee's leadership. SMALL provided a structured language suitable for teaching compiler construction on limited-capacity machines. Its design reflected contemporary pedagogical debates about appropriate introductory languages: while some advocated BASIC for accessibility, others preferred the discipline of Algol-style syntax. SMALL thus represents an early example of language design adapted to local educational and hardware contexts.

Kiwinet and Early Networking

In 1975, staff at Massey University and Victoria University of Wellington established Kiwinet, a pilot project for sharing resources between their B6700 systems. The network achieved basic connectivity but lacked the software infrastructure required for sustained practical use. Nevertheless, it provided valuable experience in inter-institutional data communication and anticipated later developments in New Zealand's research networking [37]. Full Internet connectivity would not arrive until 1989.

Legacy

The early decades of computing in New Zealand were defined by adaptation and experimentation. Limited resources compelled technical creativity, and the modifications made to imported systems were often as significant as the machines themselves. Some of these early systems—such as the IBM 1620 and 1130—survive today in the Bob Doran Museum of Computing at the University of Auckland.

ECONOMIC IMPACT

In 1960, the New Zealand economy was very tightly coupled to the UK, despite being literally a world away. The New Zealand currency was held at par with the British pound, and the UK accounted for 53% of exports and 44% of imports. Moreover, 90% of the total value of exports was derived from agricultural produce [1]. However, as early as 1961, Britain announced its interest in joining the European Economic Community (EEC), and this was correctly interpreted in New Zealand as a serious warning that access to the UK market would sooner or later become difficult.

In fact, diversification had already started, with meat exports going to 60 different countries. There was a successful effort at further diversification of both exports and imports. For example, tariff reductions were negotiated with Japan, and a free trade agreement with Australia was signed in 1965 [38]. Although New Zealand's access to UK markets was eased by the so-called Luxembourg agreement in 1971, when Britain eventually joined the EEC in 1973, New Zealand had already become a completely independent trading nation, as it remains today. As noted above, IBM and other American computer companies took advantage of this opportunity. By 1975 exports to the UK were down to 22% of the total and imports to 19% [13].

Thus the growth of computing in New Zealand took place against a background of major economic upheaval. This makes it particularly difficult to quantify the economic effects of computing. It is safe to assume that the companies and other organizations that increasingly used computers saw a financial advantage in doing so. As noted above, project planning (PERT) was a widespread application, which suggests that companies saw much more to computing than savings on clerical staff. Writing of government computers in 1985, A. C. Shailes [41] said "The use of computers has released staff from mundane administrative support tasks and allowed them to become directly involved in achievement of the government's main objectives. It is clear that direct financial savings from computers have recovered the overall cost of all computer resources many times over." On the banking front, Ian Archibald [19] wrote of Databank that "co-operation in basic processing systems has freed up staff to intensify competition in developing and marketing the banking products." Quantifying these effects in millions of dollars seems impossible—indeed, economists have largely failed to do so, even for intensively studied economies such as the USA [39], [42]. However, we conclude that the kind of competitive and flexible economy that New Zealand needed to survive as a trading nation by 1975 would have been essentially unachievable had it lagged in computing.

SOCIAL IMPACT

Perhaps it is no surprise that lawyers were among the first to consider the social impact of computers. A first legal analysis was published in 1971 by Francis M. Auburn [20], in which consideration of privacy was the major theme. The author observed that "there is a fear of computers and their operators among the general public" and quoted a rather naive counter-argument that "invasions of privacy are the result of misuses of the data retrieved and not a function of the storage medium" (i.e., blame the users, not the technology). With hundreds of thousands of banking transactions per day already taking place, and with a national police computer and computerized hospital records in the offing, the risks of accidental or intentional privacy breaches already seemed very real, more than 50 years ago. The legal situation was murky: "If there is a general right of privacy we are still no nearer to a definition of the circumstances in which it is applicable to computers." All sorts of data raised thorny privacy issues once computerized, such as medical records, census returns, and credit history. The issue of a unique national identity number to tie all records together was also of concern, with inevitable references to George Orwell's novel *1984*. Auburn commented that "The most rigorous presently known technological safeguards would be quite insufficient to prevent abuse." Auburn's conclusion was that a "comprehensive review of the present and future impact of electronic data processing on society" was needed, to be followed by legislation.

Auburn was not alone in raising these concerns. For example, also in 1971, A. L. C. Humphreys of ICL, the Christchurch *Press*, and the New Zealand Computer Society were all concerned about privacy. Certain lawyers even drafted a proposed law [6] and a Bill was introduced to Parliament in 1975 [11] but never made it into law. New Zealand's first Privacy Act was not passed until 1993, prior to which common law applied.

Despite this legal vacuum, plans for a national police database were made from 1972, enabled by Act of Parliament. Because of ethical concerns, the NZCS was highly involved in setting up the 1976 Wanganui Computer Centre Act. The system, officially the National Law Enforcement Data Base, often called the Wanganui Computer because it was installed in the town now known as Whanganui, was operational by 1976 (Figure 8). This led some years later to protest actions. However, it seems that during our study period, despite the concerns of some lawyers, there was little public perception of computers as a societal threat prior to the police database.

FIGURE 8. Wanganui Computer Centre, 1975. Courtesy Fletcher Archives, NZ.

Apart from privacy issues, it is hard to separate the impact of computers from other factors that changed society in the 1960s and early 1970s, as the first generation born after World War II grew to maturity. Colin Beardon [21] discussed the computerization of "women's work" and its possible impact on unemployment. Certainly, skilled women who operated ledger machines and comptometers found themselves out of a job as computers took over (Figure 9). Some of them no doubt moved into computer operations. However, many of the replacement jobs would have been less skilled and lower paid data entry jobs. Beardon noted that the fraction of the "unemployed who are female" increased from 10.9% in 1961 to 18.0% in 1971 and 34.9% in 1975, but is impossible to know how much of this change was caused by computers. For example, in retail banking, according to anecdotal evidence, the technological change from manual to computerized operations changed people's jobs, but did not destroy them [33]. The loss of typing and secretarial jobs came a decade later, with the advent of personal computers. Throughout our period of study, computers were mainly hidden from public view except for people directly involved, and had relatively little direct social impact compared to other modern technology such as television. Apart from the job market, where computers had a direct effect on some working lives, they were not even a talking point for most people.

Initial reactions to the introduction of computers were therefore based on curiosity. The first large mainframes were used in only a few large organizations, and had little impact on the general public. During this period employment was high and although there was some discussion about whether computers would result in mass unemployment, generally they were viewed positively and there was optimistic discussion about shorter working weeks and an improved quality of life. One concern was whether computers would improve individual autonomy or lead to increased conformity and control. Speaking in 1968, Databank's General Manager, Gordon Hogg, felt that automation would fit well with New Zealand values since a reduction in unskilled routine jobs would create a levelling in occupational social distinction, along with a continuing increase in income, both favored in a country committed to social equality (citation???). This reflected his experience at Databank where the new computer system did not lead to any job losses but rather released staff from routine functions like ledger posting and calculating interest so they were able to provide a more personal service to customers.

The NZCS did take on the responsibility of considering the impact that computers would have on broader society. The third conference in 1972 had a theme of 'Computers in the Community' and aimed to ensure that in New Zealand computers continued to be used for the greatest public benefit and the least cost (citation???). The Computer Society's president, Perce Harpham, remarked that this conference "marks the maturing of our profession. We can no longer ignore the fact that our work affects the lives of many of our fellows and in turn that we are dependent on them" [28]. The fifth national conference in 1976 also considered social impact, the theme was 'Responsibility'. Keynote speaker, Paul Sieghart, from the UK Data Protection Committee stressed the responsibility that workers in the computing field had to educate the general public. Education is key, he said, as most people are ignorant about computing, regarding it "as a form of magic – if not sorcery". The computer professional's job was to demystify the computer in order to encourage the widest and most open public debate about all aspects of the social consequences of the developing technologies [43].

FIGURE 9. BNZ Ledger Room, 1950s. Courtesy BNZ Museum.

CONCLUSION

We have described how computers penetrated government, business, tertiary education and research throughout New Zealand between 1960 and 1975. This was the period when the country progressed from a post-colonial society strongly linked to the UK to become a self-sufficient trading nation with its own strong identity. Modern computing, local innovation and entrepreneurship combined to support the needs of the evolving economy. Without them, it is unlikely that New Zealand could have succeeded as it did. At the end of the period, computing was entering the public consciousness as a technology that would soon have profound social implications, although personal computing and social networking lay far in the future.

The ongoing story after 1975 has been well documented by ITPNZ [46] and by InternetNZ [37].

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